

Cisco ASAv Firewall & Security Zones Lab

Network Engineering & Security Portfolio

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1 Overview

This lab report documents the configuration and verification of a Cisco ASAv firewall protecting an internal network and a DMZ. The objective is to establish security zone interfaces with appropriate security levels, enable Dynamic NAT for outbound internet connectivity, and verify HTTP access from internal clients to both external internet hosts and DMZ resources.

2 Lab Prerequisites & Topology

- Cisco ASAv 9.8.1 appliance configured with three security zones.
- Layer 2 switches providing isolation via dedicated VLANs (VLAN 100 for Inside, VLAN 200 for DMZ).
- Internal clients (`webterm-1`, `PC2`) and DMZ services (`Web-Server`, `Mail-Server`).
- Outside connectivity provided via DHCP on the external interface pointing to a NAT gateway.

2.1 Addressing & Security Zone Schema

3 Step-by-Step Configuration Guide

3.1 Step 1: ASAv Interface Security Zones

Configure interface naming, security levels, and IP address bindings across all three interfaces.

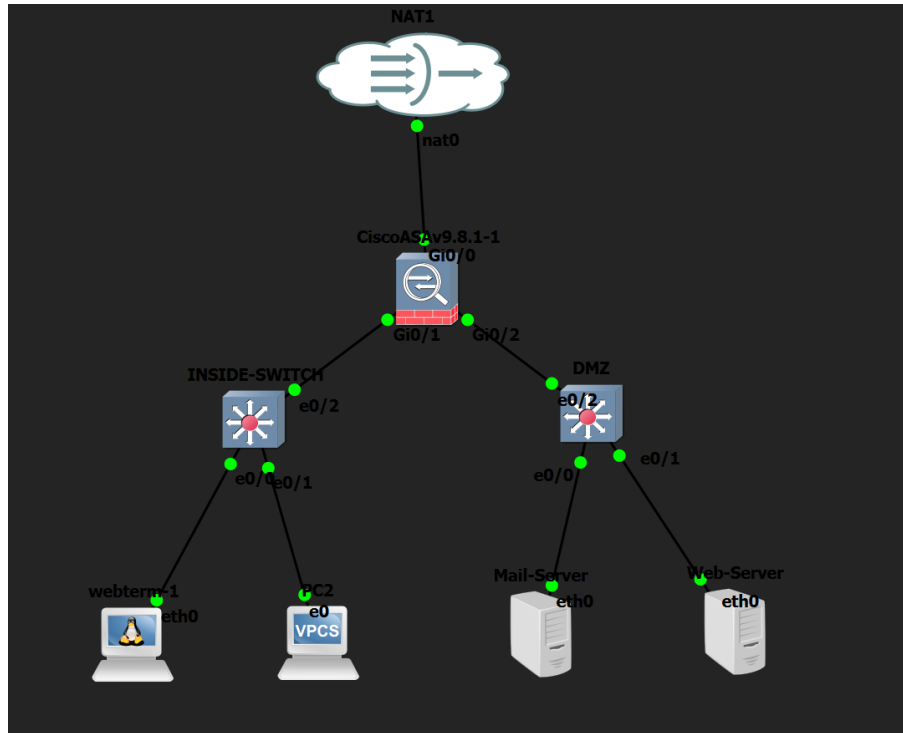


Figure 1: Cisco ASA Network Topology Diagram.

Device / Interface	Zone Name	Security Level	IP Address / Subnet	VLAN
ASA GigabitEthernet0/0	outside	0	DHCP (Dynamic)	N/A
ASA GigabitEthernet0/1	inside	100	192.168.10.1 /24	VLAN 100
ASA GigabitEthernet0/2	dmz	50	172.16.10.1 /24	VLAN 200
Inside-Switch	INSIDE-NETWORK	N/A	N/A	VLAN 100
DMZ-Switch	DMZ	N/A	N/A	VLAN 200
Web-Server	DMZ	N/A	172.16.10.100 /24	VLAN 200

Table 1: Addressing and Security Zone Mapping.

Listing 1: ASA Interface Configuration

```

interface GigabitEthernet0/0
  nameif outside
  security-level 0
  ip address dhcp
!
interface GigabitEthernet0/1
  nameif inside
  security-level 100
  ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/2
  nameif dmz
  security-level 50
  ip address 172.16.10.1 255.255.255.0

```

3.2 Step 2: Switch Access Layer VLAN Configuration

Assign access ports on both the Inside and DMZ switches to isolate local broadcast domains.

INSIDE-SWITCH Configuration (VLAN 100):

```
INSIDE-SWITCH# show vlan brief
100  INSIDE-NETWORK                active    Et0/0, Et0/1, Et0/2

INSIDE-SWITCH# show running-config interface ethernet 0/0
interface Ethernet0/0
  switchport access vlan 100
  switchport mode access
```

DMZ-SWITCH Configuration (VLAN 200):

```
DMZ-Switch# show vlan brief
200  DMZ                          active    Et0/0, Et0/1, Et0/2

DMZ-Switch# show running-config interface ethernet 0/0
interface Ethernet0/0
  switchport access vlan 200
  switchport mode access
```

3.3 Step 3: Dynamic NAT Configuration

Configure Network Address Translation to permit outbound traffic from both the **inside** and **dmz** zones through the external interface.

Listing 2: ASA v Network Objects and Dynamic NAT

```
object network OBJ_INSIDE_NET
  subnet 192.168.10.0 255.255.255.0
  nat (inside,outside) dynamic interface

object network OBJ_DMZ_NET
  subnet 172.16.10.0 255.255.255.0
  nat (dmz,outside) dynamic interface
```

4 Verification & Connectivity Testing

4.1 Step 4: End-to-End Traffic Verification

Test HTTP connectivity from an internal client (**webterm-1**) to the local DMZ web server as well as external internet destinations.

Listing 3: webterm-1 Terminal Verification Output

```
root@webterm-1:~# curl -I http://172.16.10.100
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.14.4
Date: Sun, 27 Sep 2026 22:09:01 GMT
Content-type: text/html; charset=utf-8
Content-Length: 457

root@webterm-1:~# curl -I "http://example.com"
HTTP/1.1 200 OK
Date: Sun, 27 Sep 2026 22:09:06 GMT
Content-Type: text/html
Connection: keep-alive
Server: cloudflare
cf-cache-status: HIT
```

- **Inside to DMZ (100 → 50):** Higher-to-lower security level traffic is permitted by default.
- **Inside to Outside (100 → 0):** Dynamic NAT successfully translates the internal address onto the outside interface for internet reachability.